

Comprehensive Quantum SAC Analysis Report

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RANDOM DATASETS (10 datasets):

- Average Accuracy: 0.6770 ± 0.1097
- Best Accuracy: 0.8596
- Average Training Time: 11.81s

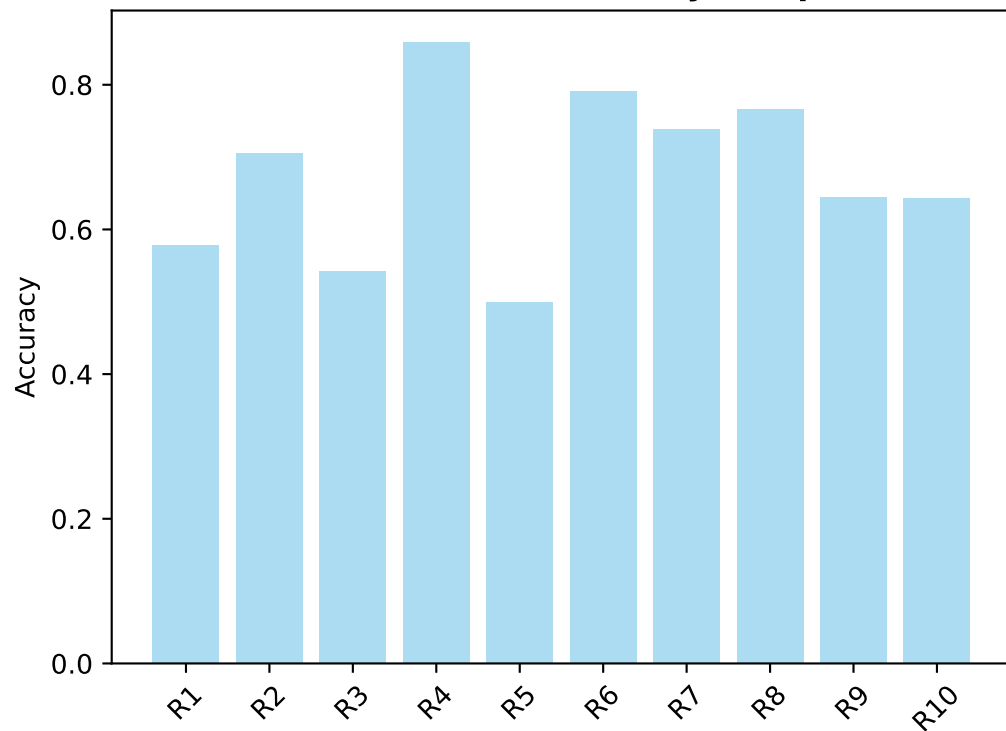
EARTHQUAKE DATASETS (4 sample sizes):

- Average Accuracy: 0.9854 ± 0.0036
- Best Accuracy: 0.9900
- Average Training Time: 51.47s

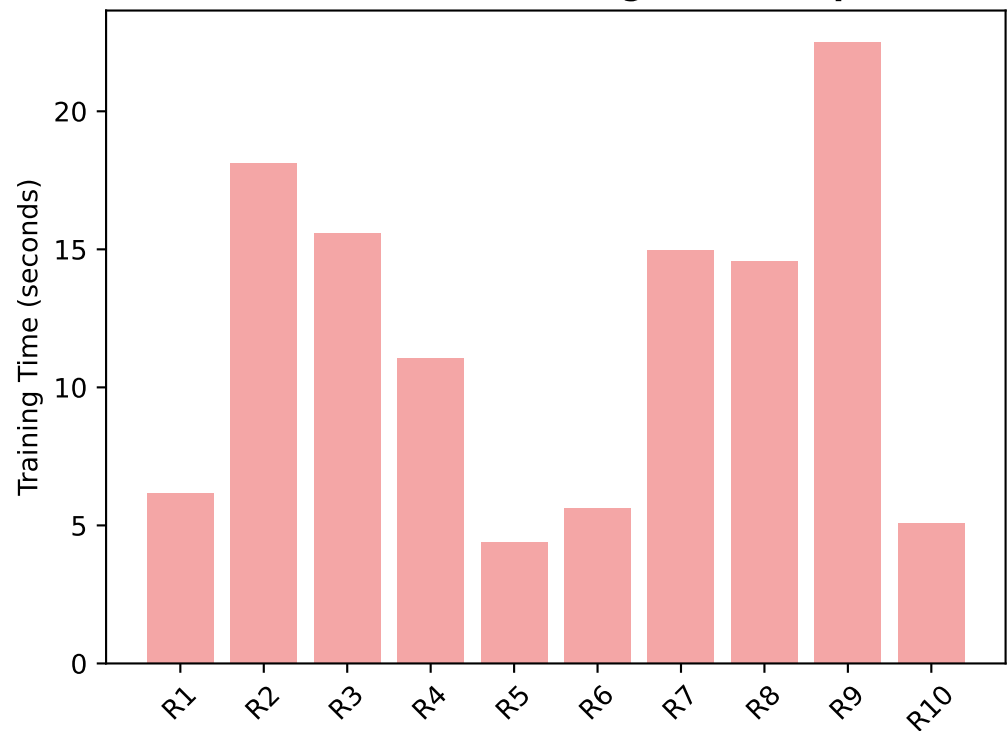
BEST MODELS:

- Best Random: Dataset #4 - 0.8596 accuracy
- Best Earthquake: earthquake_500 - 0.9900 accuracy

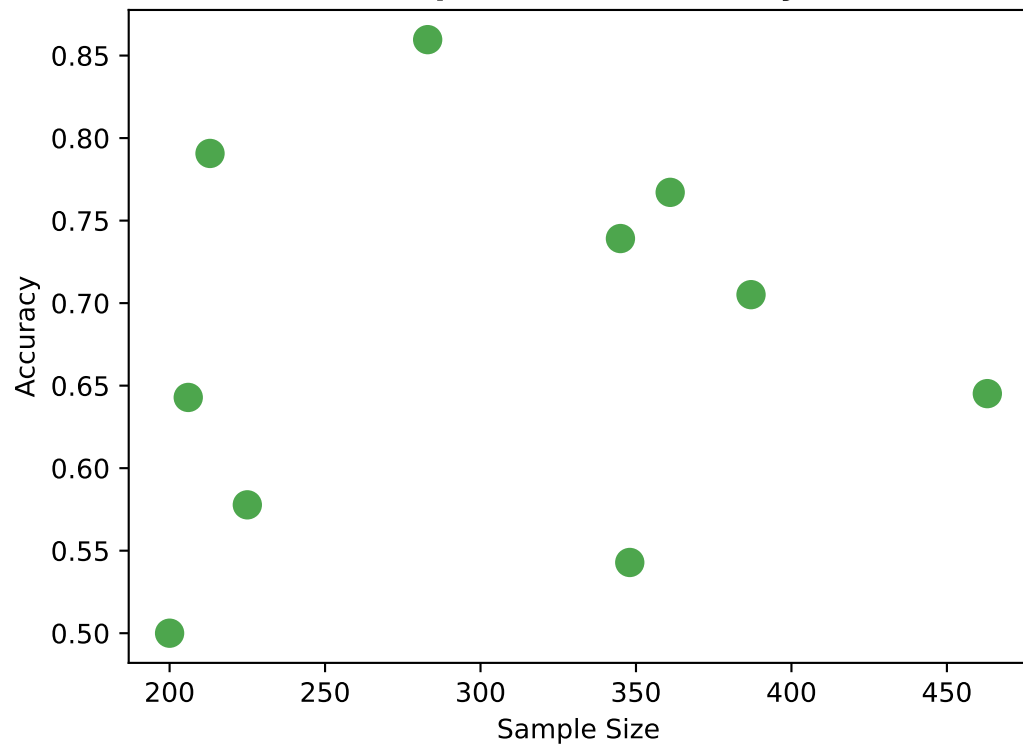
Random Datasets - Accuracy Comparison



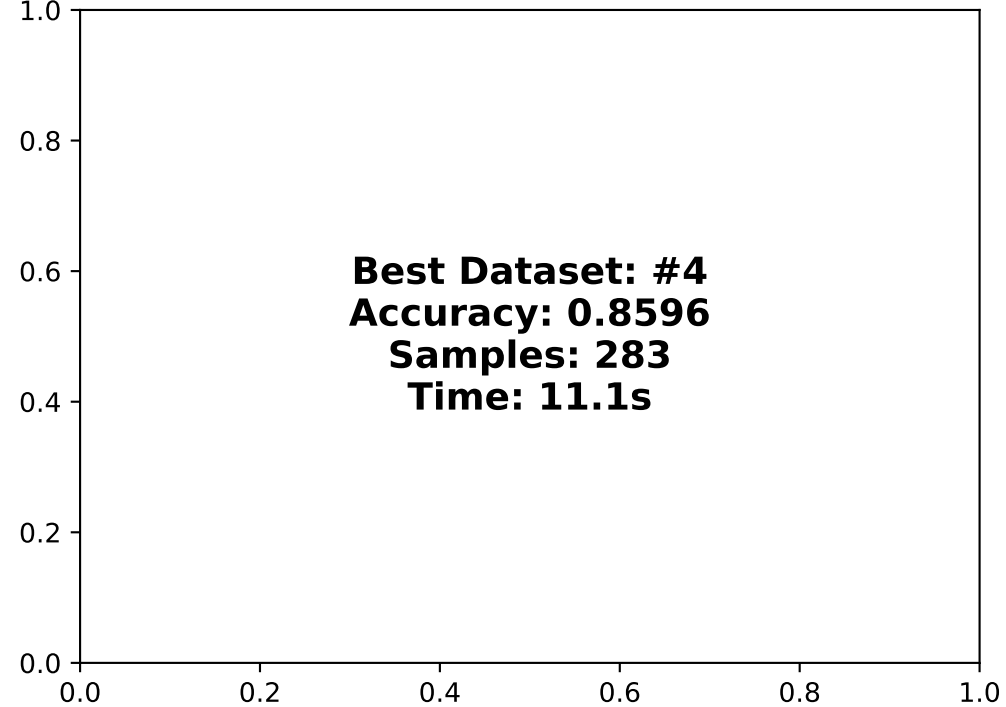
Random Datasets - Training Time Comparison



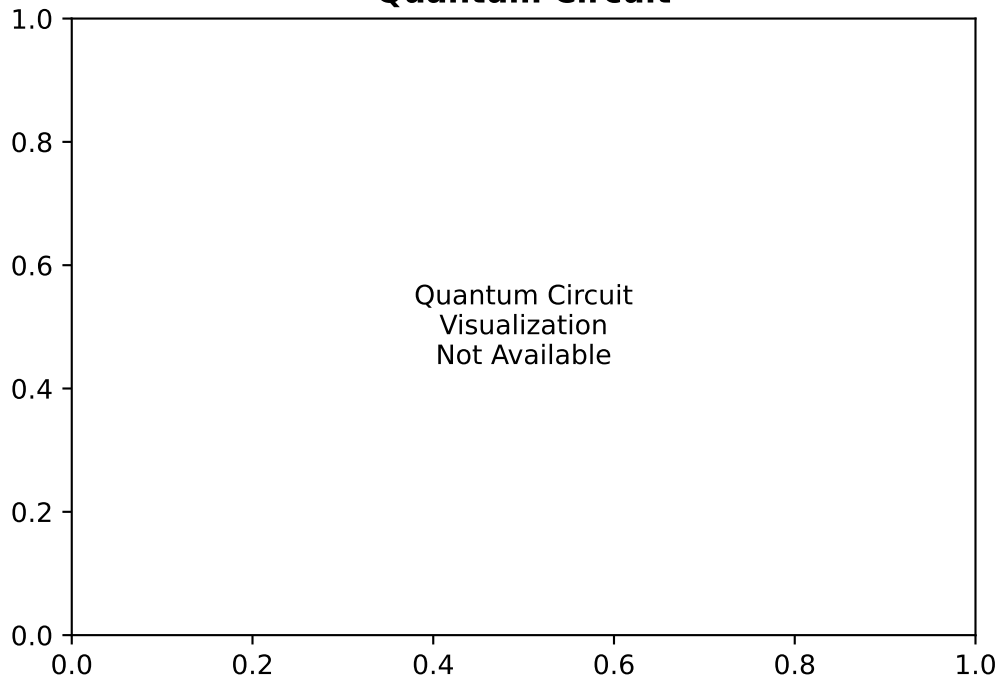
Sample Size vs Accuracy



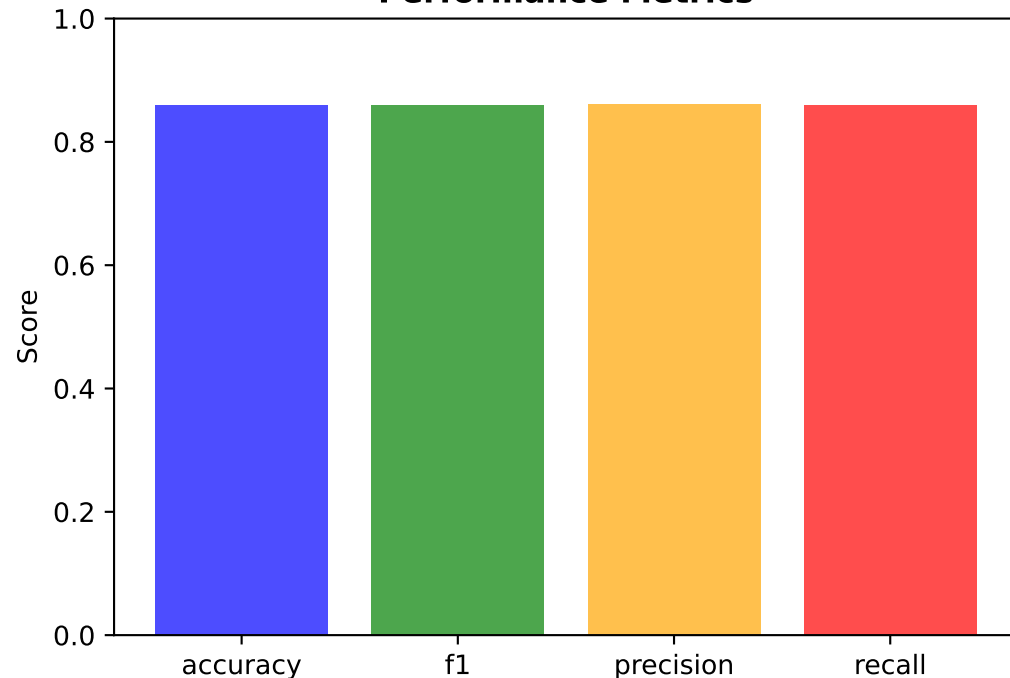
Best Random Dataset



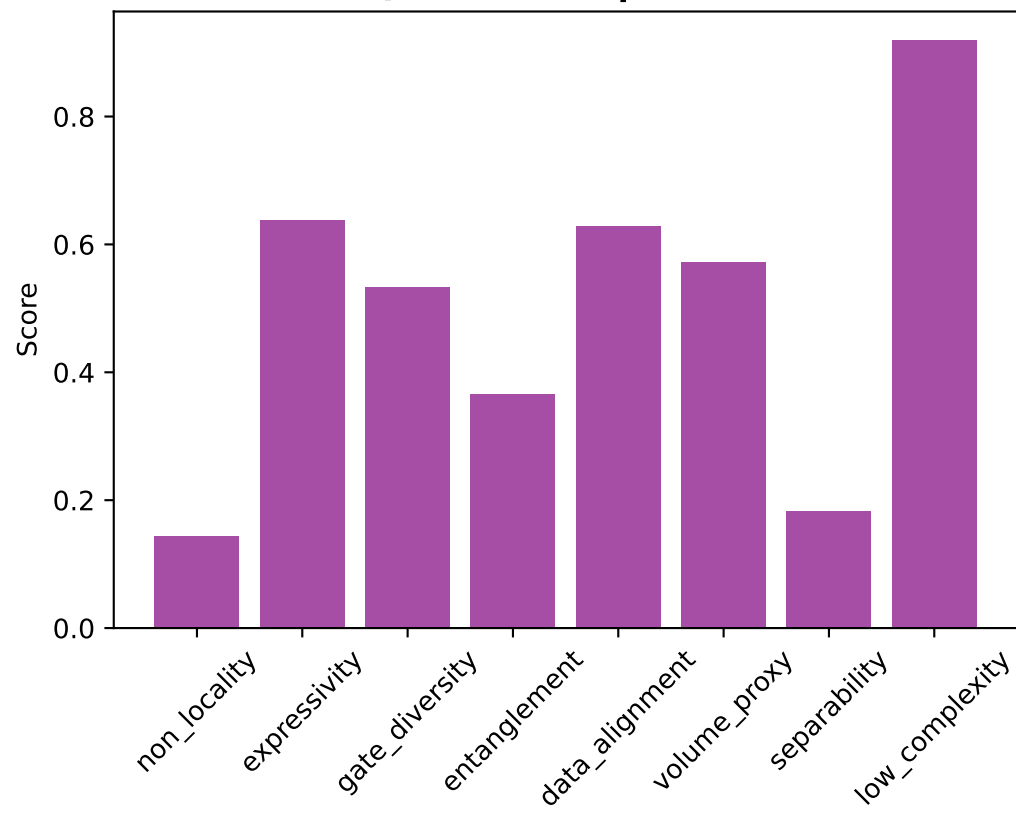
Quantum Circuit



Performance Metrics



QMetrics Components



Dataset & Training Info

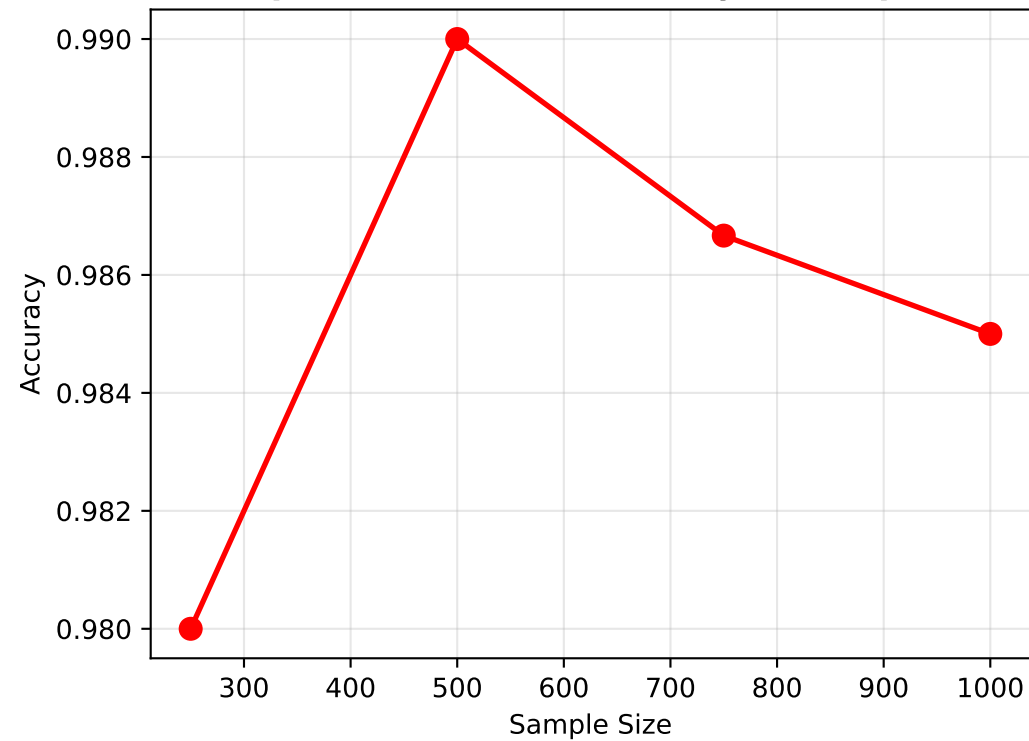
Dataset Information:

- ID: #4
- Samples: 283
- Features: 6
- Class Distribution: [142, 141]

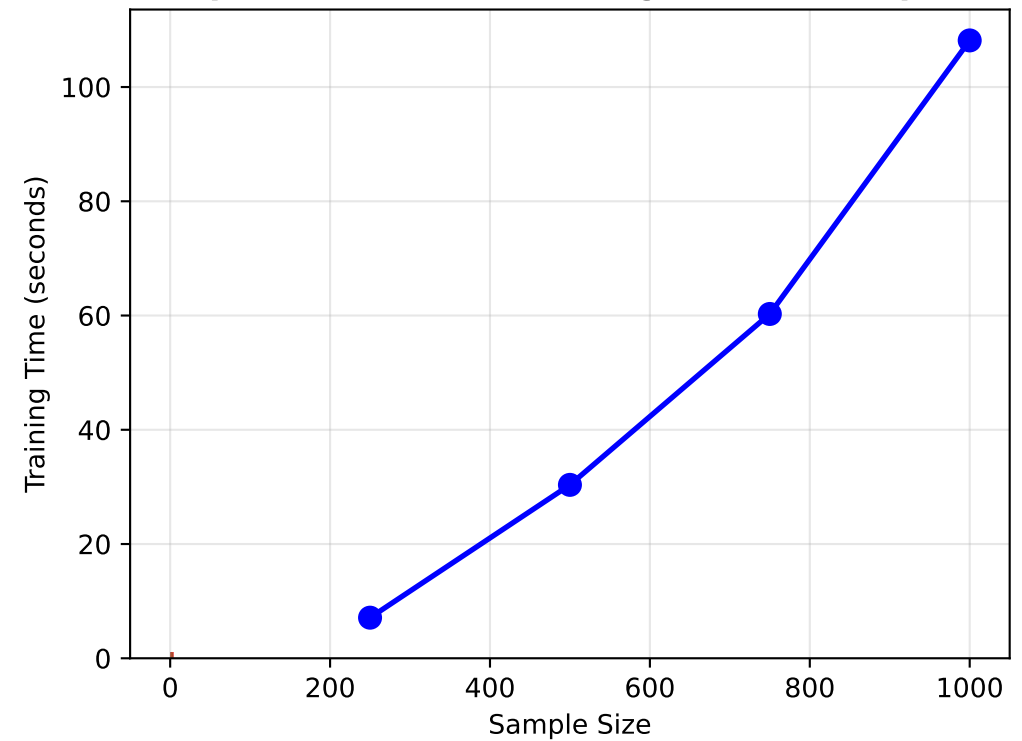
Training Performance:

- SAC Training Time: 0.05s
- QSVC Training Time: 11.01s
- Total Time: 11.06s
- Best Reward: 0.4402
- Average Reward: 0.3841

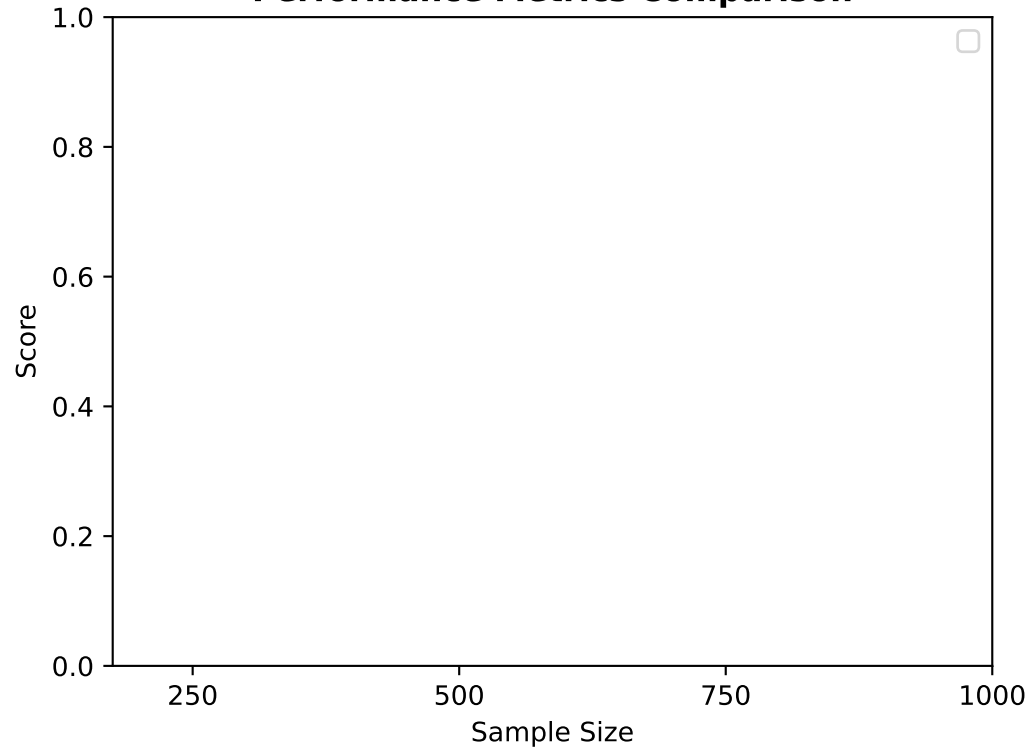
Earthquake Datasets - Accuracy vs Sample Size



Earthquake Datasets - Training Time vs Sample Size



Performance Metrics Comparison

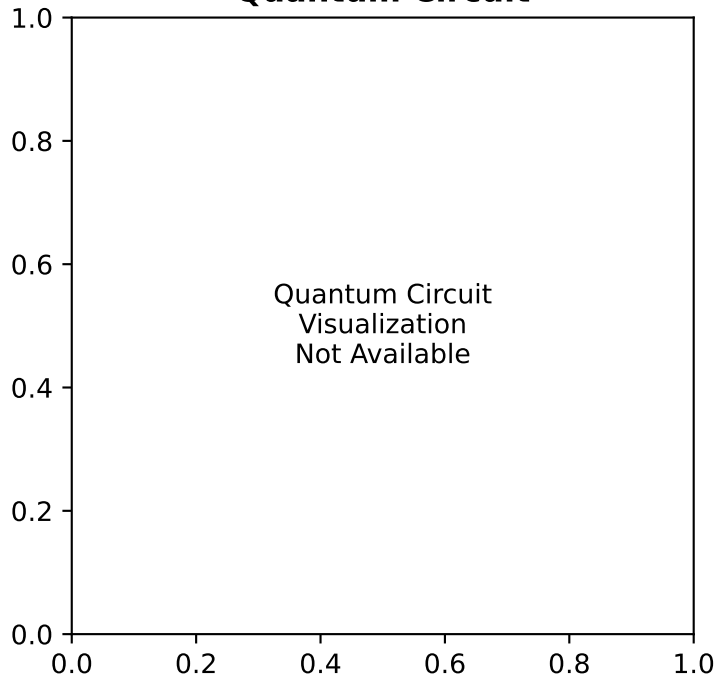


Feature Selection Results

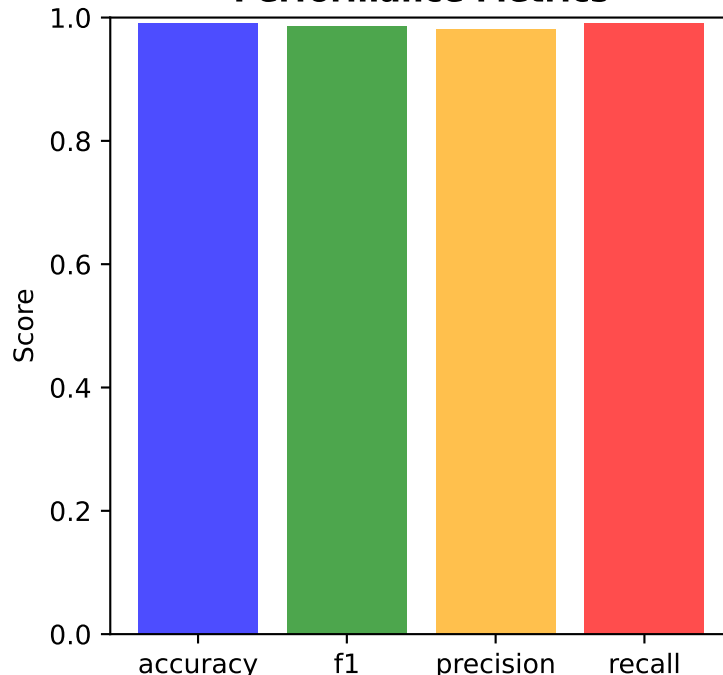
Selected Features (Feature Selection):

1. latitude
2. longitude
3. depth
4. gap
5. rms
6. magType_encoded

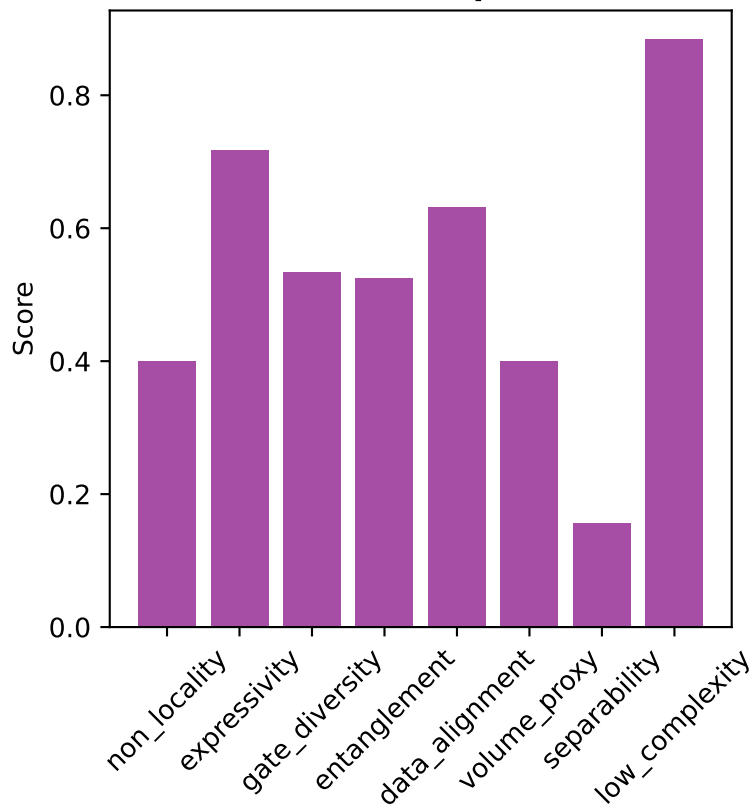
Quantum Circuit



Performance Metrics



QMetrics Components



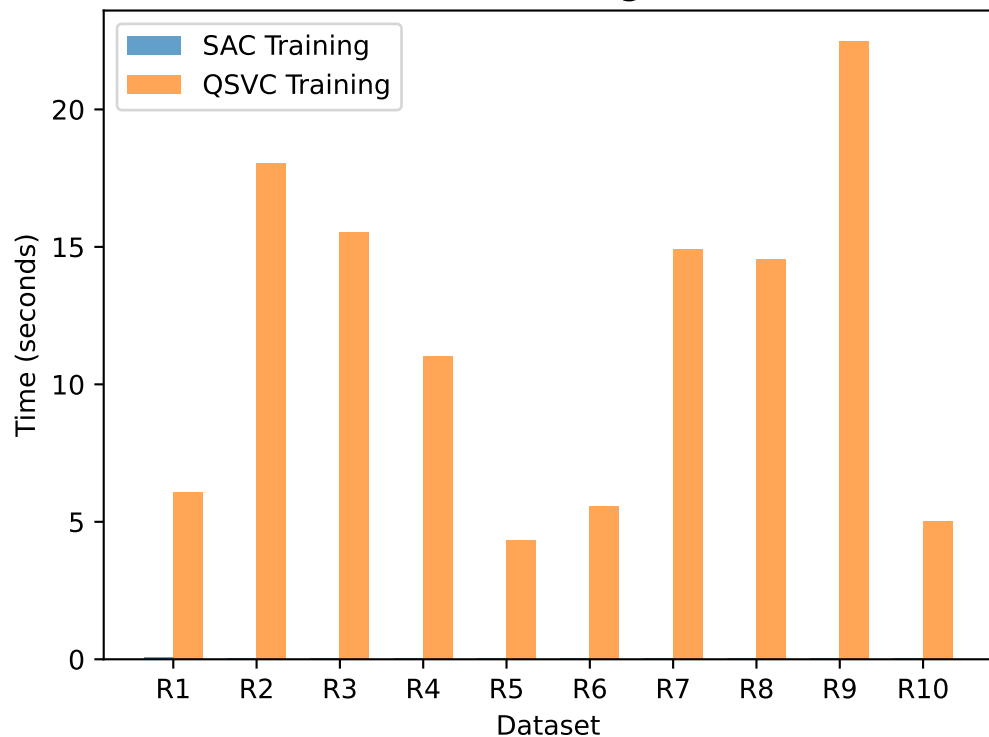
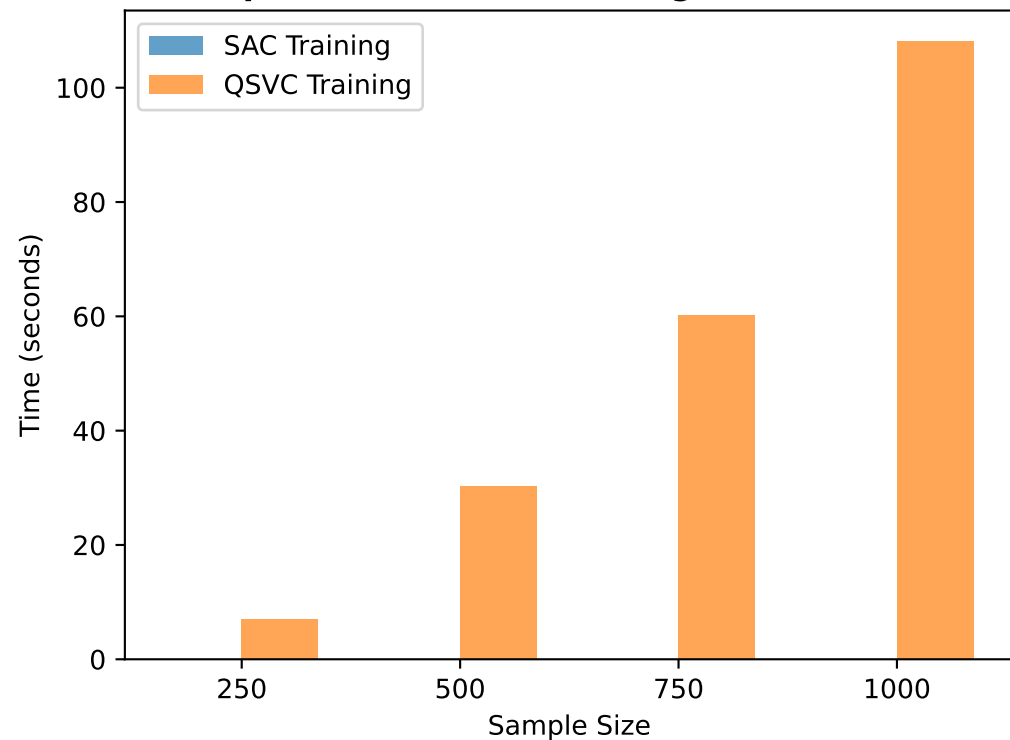
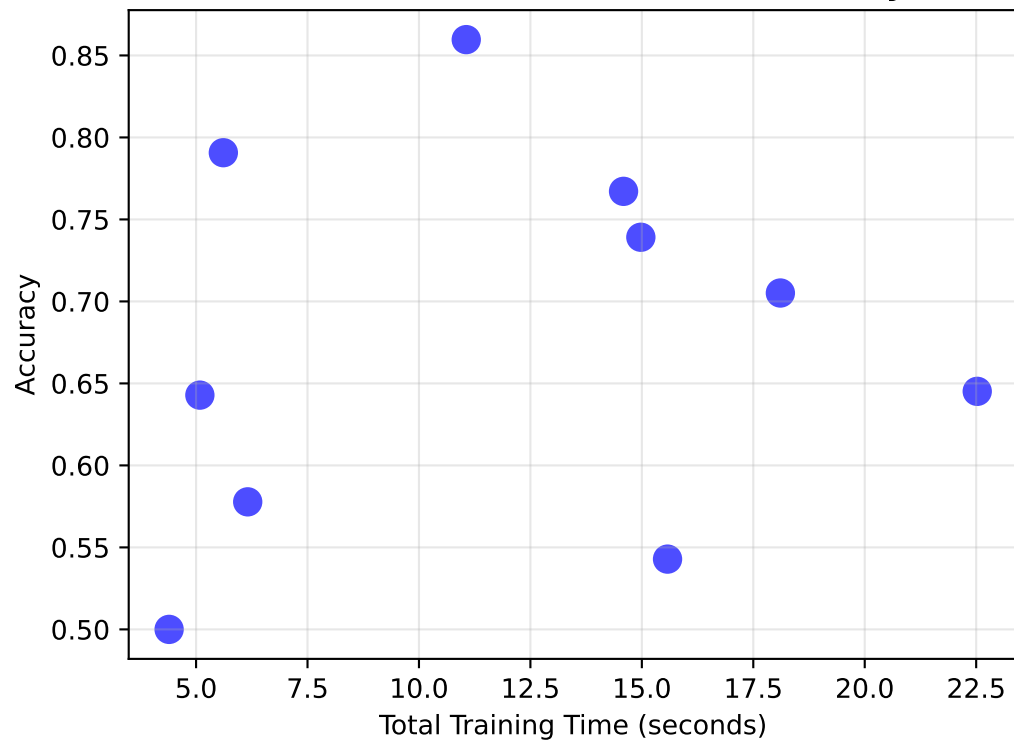
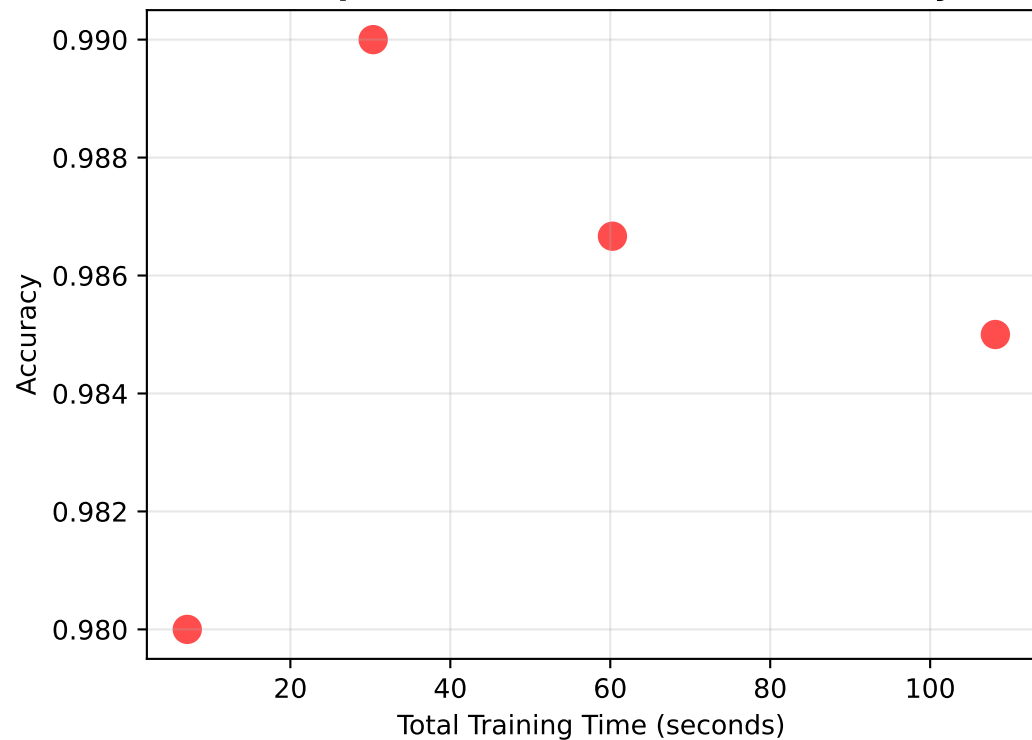
Dataset & Training Info

Dataset Information:

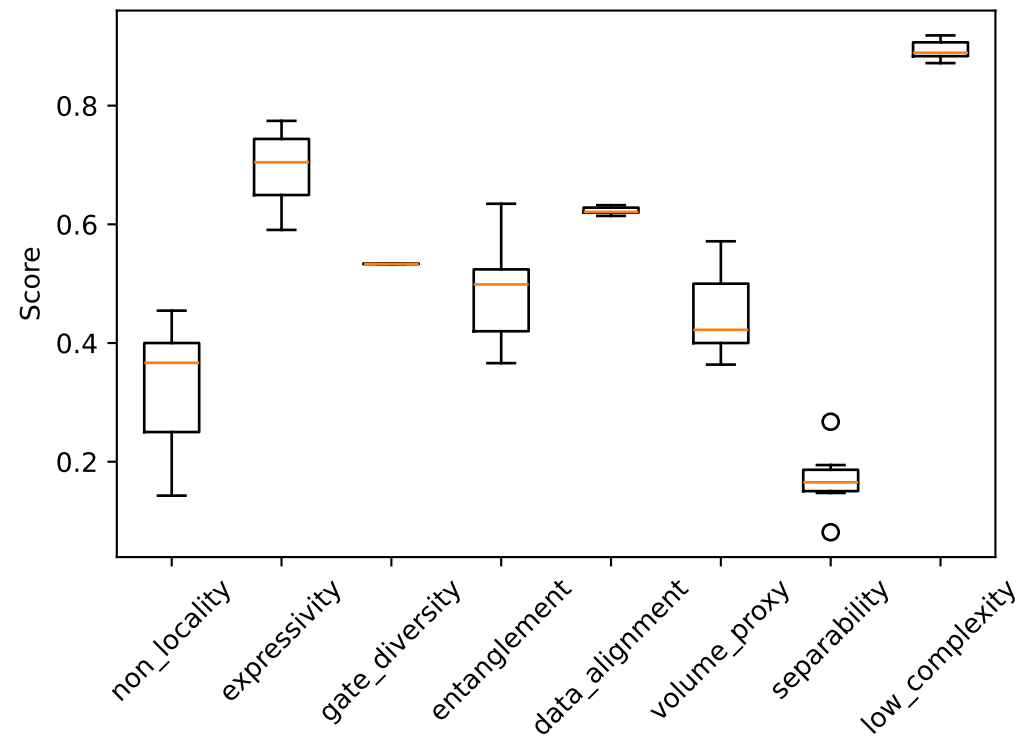
- Sample Size: 500
- Features: ['latitude', 'longitude', 'depth', 'gap', 'rms', 'magType_encoded']
- Class Distribution: [493, 7]

Training Performance:

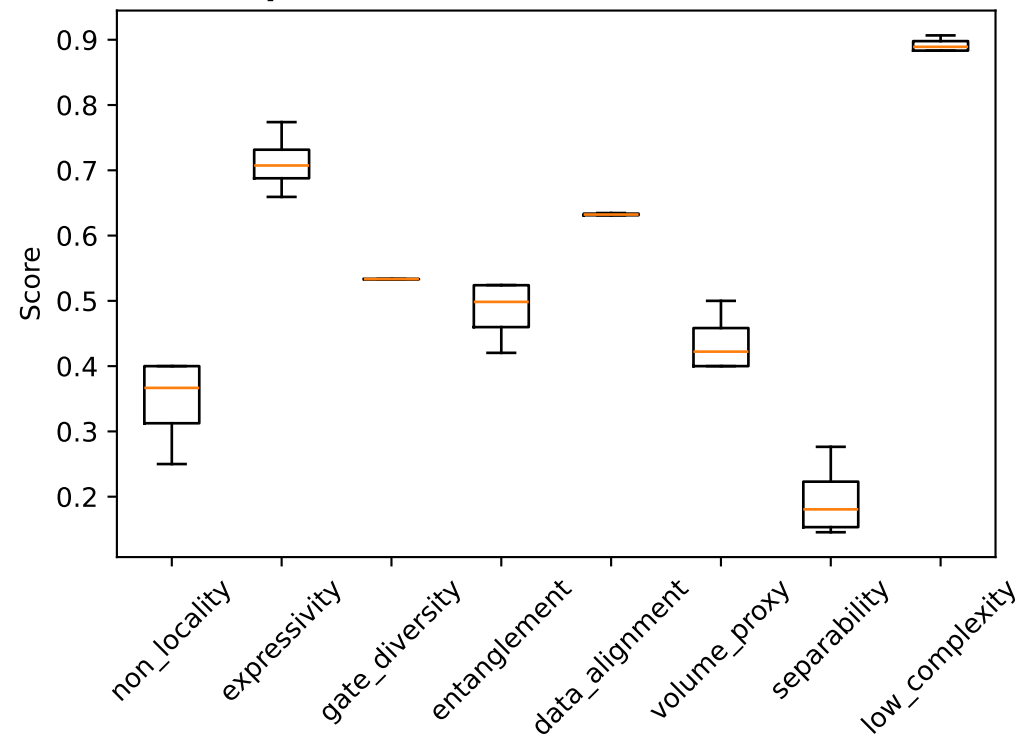
- SAC Training Time: 0.05s
- QSVC Training Time: 30.31s
- Total Time: 30.36s
- Best Reward: 0.4224
- Average Reward: 0.3891

Random Datasets - Training Time Breakdown**Earthquake Datasets - Training Time Breakdown****Random Datasets - Time vs Accuracy****Earthquake Datasets - Time vs Accuracy**

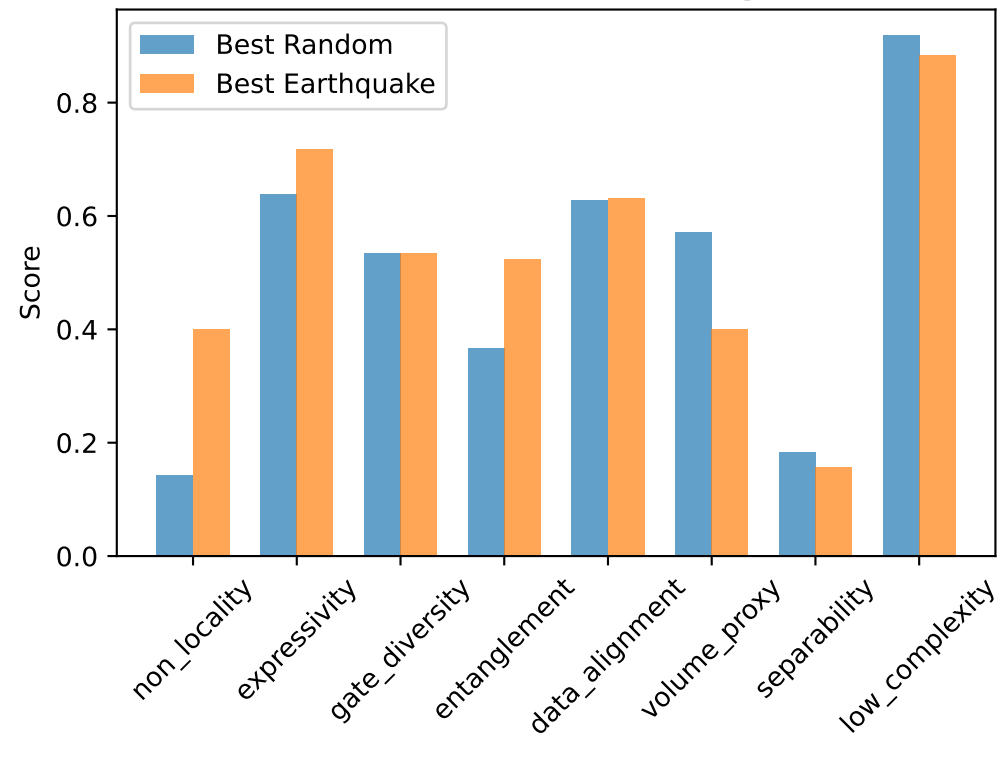
Random Datasets - QMetrics Distribution



Earthquake Datasets - QMetrics Distribution



Best Models - QMetrics Comparison



QMetrics vs Accuracy Correlation

